

Business Case

A client faced a critical operational issue where their traditional download-first workflow for handling external data introduced storage overhead, transfer failure risks, and unnecessary infrastructure costs. More concerning was the possibility of losing valuable data despite the company's significant investment in digital systems, infrastructure maintenance, and internal IT operations. This required immediate resolution to prevent future losses.

Although the company's technical environment was well-funded and actively maintained, the company preferred the IT team to handle their data for security reasons. However, certain data issues could not be immediately identified by the IT team. After several attempts to resolve the problem, it became clear that the challenge was not purely infrastructural—it was deeply tied to data movement, reliability, and pipeline design. As a result, the organization recommended bringing in a data professional to lead the resolution.

Project Solution

As the data professional engaged on the project, I validated that cloud storage was only one part of the solution. The larger need was to redesign how data moved across the system.

The problem was that critical external datasets needed to be transferred into cloud storage directly from the ERP and CRM systems, reliably, and without dependence on local storage systems. So, I designed a direct-to-cloud streaming pipeline that moved data from source systems straight into cloud storage, eliminating local persistence while improving operational efficiency, reliability, and scalability.